

Key Answers Final Exam

20004E 280824 (AUG 2024)

1. a. Microcontroller :

- stronger computing capacity for analytical control
- Can be found often in single board configuration

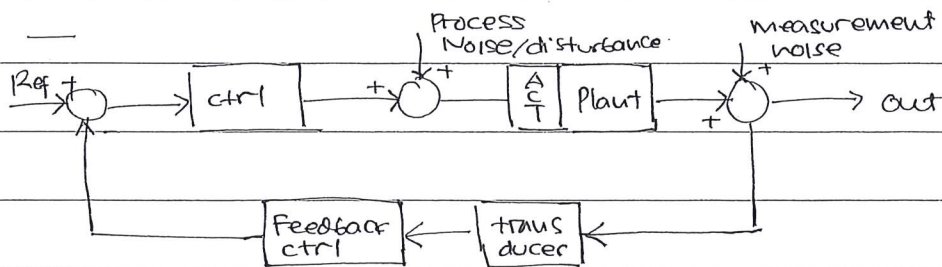
PLC :

- used more often for logic or timer based or batch process control
- Found in most cases, if not all, in sturdy casing form

b. Control system example using

- Microcontroller : drone control system, mobile robot, etc.
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- PLC : traffic light, lift/escalator, etc.

c. Feedback control system block diagram :



Necessary components:

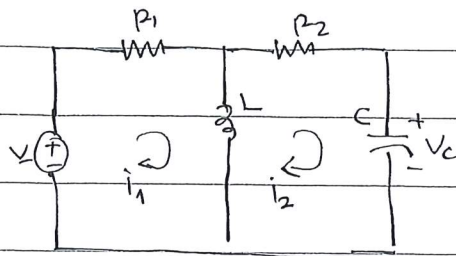
Plant + actuator, controller (either FF or FB), sensor/transducer.

The sensor can be excluded from open-loop control

as there is no need to return the output to the loop.

2.

RLC circuit:



$$R_1 = 1 \Omega$$

$$C = 1 \text{ F}$$

 $V(t)$ input

$$R_2 = 1 \Omega$$

$$L = 1 \text{ H}$$

 $V_c(t)$ output

System's dynamic equations:

$$(R_1 + Ls) i_1(s) - Ls i_2(s) = V(s) \quad \dots (1)$$

$$(R_2 + Ls + \frac{1}{Cs}) i_2(s) - Ls i_1(s) = 0 \quad \dots (2)$$

$$V_c(s) = \frac{1}{Cs} i_2(s) \quad \dots (3)$$

a. Transfer function $\frac{V_c(s)}{V(s)}$ From (2) we want to separate i_1

$$i_1(s) = \frac{R_2 + Ls + \frac{1}{Cs}}{Ls} i_2(s) \quad \dots (4)$$

From (3), we want to separate i_2

$$i_2(s) = Cs V_c(s) \quad \dots (5)$$

Substitute (4) to (1), then substitute (5) $\Rightarrow \left\{ \frac{(1+s)(1+s+\frac{1}{s}) - s^2}{s} \right\} s V_c = V$

$$(R_1 + Ls) i_1(s) - Ls i_2(s) = V_c(s)$$

$$(s^2 + 2s + 2 + \frac{1}{s} - s^2) V_c = V$$

$$(R_1 + Ls) \frac{R_2 + Ls + \frac{1}{Cs}}{Ls} i_2 - Ls i_2 = V$$

$$\frac{V_c}{V} = \frac{1}{2s + 2 + \frac{1}{s}}$$

$$\left\{ \frac{(R_1 + Ls)(R_2 + Ls + \frac{1}{Cs}) - Ls}{Ls} \right\} i_2 = V$$

$$= \frac{s}{2s^2 + 2s + 1}$$

$$\left\{ (R_1 + Ls) \frac{(R_2 + Ls + \frac{1}{Cs})}{Ls} - Ls \right\} Cs V_c = V$$

Substitute the value of the parameters:

$$\left\{ \frac{(1+s)(1+s+\frac{1}{s})}{s} - s \right\} s V_c = V$$

b Laplace inverse of $V_c(s)$ if $V_c(s) = \frac{1}{s}$ (unit step)

$$\frac{V_c(s)}{V(s)} = \frac{s}{2s^2 + 2s + 1}$$

$$V_c(s) = \frac{s}{2s^2 + 2s + 1} \quad V(s) = \frac{s}{2s^2 + 2s + 1} \cdot \frac{1}{s} = \frac{1}{2s^2 + 2s + 1}$$

$$= \frac{1/2}{s^2 + s + \frac{1}{2}} = \frac{1/2}{(s + \frac{1}{2})^2 + \frac{1}{4}} = \frac{1/2}{(s + \frac{1}{2})^2 + (\frac{1}{2})^2}$$

Using Laplace table

$$V_c(t) = \mathcal{L}^{-1}(V_c(s)) = e^{-\frac{1}{2}t} \sin(\frac{1}{2}t)$$

c Final value or steady state value of $V_c(t)$ with $V(t)$ unit step

using FVT :

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

$$\lim_{t \rightarrow \infty} V_c(t) = \lim_{s \rightarrow 0} s V_c(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{2s^2 + 2s + 1} = 0$$

d. Using (b)

$$V_{css} = \lim_{t \rightarrow \infty} V_c(t) = \lim_{t \rightarrow \infty} e^{-\frac{1}{2}t} \sin(\frac{1}{2}t) = 0, \text{ as } e^{-\frac{1}{2}t} \rightarrow 0, t \rightarrow \infty$$

because as $t \rightarrow \infty$, $e^{-\frac{1}{2}t} \rightarrow 0$

3. A shaker table as in given picture with

M : mass of table

m : mass of camera

K_s : spring stiffness constant

B : damping constant

input force: $f_i(t)$, output: table displacement

Equation of motion of the table:

$$a\ddot{x} + b\dot{x} + cx = u(t)$$

$$a. (M+m)\ddot{x} + B\dot{x} + K_s x = f_i$$

input = f_i (coil force)

output: x (displacement of table)

b. The state space model

Define: $x_1 = x$ output: x

$$x_2 = \dot{x}$$

$$u = f_i$$

Hence:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \ddot{x} = \frac{-K_s x - B\dot{x} + f_i}{M+m} = \frac{-K_s x_1 - Bx_2 + u}{M+m}$$

State space model equation

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ \frac{-K_s}{M+m} & \frac{-B}{M+m} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{u}{M+m}$$

Output equation

$$y = x = x_1$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}$$

$$\therefore A = \begin{bmatrix} 0 & 1 \\ \frac{-K_s}{M+m} & \frac{-B}{M+m} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{M+m}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad D = 0$$

c $m = 250 \text{ g}$, $k_s = 1.1 \times 10^7 \text{ N/m}$

$B = 1100 \text{ N/m.s}$

Range of M for stability?

The system is stable, if eigenvalues of A are all ^{have} negative real parts.

$$A = \begin{bmatrix} 0 & 1 \\ \frac{-k_s}{M+m} & \frac{-B}{M+m} \end{bmatrix}$$

$$|sI - A| = \begin{vmatrix} s & -1 \\ \frac{k_s}{M+m} & s + \frac{B}{M+m} \end{vmatrix} = s(s + \frac{B}{M+m}) + \frac{k_s}{M+m} = 0$$

$$s^2 + \frac{B}{M+m} s + \frac{k_s}{M+m} = 0$$

$$s_{1,2} = \frac{-\frac{B}{M+m} \pm \sqrt{(\frac{B}{M+m})^2 - 4 \cdot 1 \cdot \frac{k_s}{M+m}}}{2}$$

$$= \frac{-\frac{B}{M+m} \pm \frac{1}{2(M+m)} (B^2 + 4k_s(M+m))^{1/2}}{2} < 0$$

$$= -B \pm (B^2 + 4k_s(M+m))^{1/2} < 0$$

$$B^2 - 4k_s(M+m) < B^2 \quad \text{and} \quad B^2 - 4k_s(M+m) < 0$$

$$4k_s(M+m) > 0$$

$$4k_s(M+m) > B^2$$

$$M > 0$$

$$M+m > \frac{B^2}{4k_s}$$

$$-B + (B^2 + 4k_s(M+m))^{1/2} < 0$$

$$M > \frac{B^2}{4k_s} - m$$

=

4.

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$

$$y = [1.5 \quad 0.625] x$$

a Stability : all eigenvalues have negative real parts.

$$|sI - A| = \begin{vmatrix} s+4 & 1.5 \\ -4 & s \end{vmatrix} = s(s+4) + 4(1.5) = s^2 + 4s + 6$$

$$s_{1,2} = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 1 \cdot 6}}{2} = -2 \pm \sqrt{4 - 6} = -2 \pm \sqrt{-2} = -2 \pm j\sqrt{2}$$

Real parts are negative, hence the system is stable

type of stability : asymptotic stability

b Equivalent TF

$$\frac{Y}{U} = C [sI - A]^{-1} B$$

$$= [1.5 \quad 0.625] \cdot \frac{1}{s^2 + 4s + 6} \begin{bmatrix} s & -1.5 \\ 4 & s+4 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{s^2 + 4s + 6} [1.5s + \cancel{2.5} - 2.25 + 0.625(s+4)] \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= \frac{3s + 5}{s^2 + 4s + 6} \leftarrow \text{equivalent}$$

$$\begin{aligned} \text{c. } G(s) &= \frac{3s + 5}{s^2 + 4s + 6} \Rightarrow T(s) = \frac{G(s)}{1 + G(s)} = \frac{3s + 5}{s^2 + 4s + 6 + 3s + 5} \\ &= \frac{3s + 5}{s^2 + 7s + 11} \end{aligned}$$

d. Steady state error of $G(s)$ and $T(s)$ w.r.t. unit step input
 $R(s) = \text{unit step} = \frac{1}{s}$

$$E_{ss} \text{ for } G(s) = \lim_{s \rightarrow 0} s \cdot (1 - G(s)) \cdot \frac{1}{s} R(s)$$

$$= \lim_{s \rightarrow 0} s \left(1 - \frac{3s+5}{s^2+4s+6} \right) \cdot \frac{1}{s}$$

$$= \lim_{s \rightarrow 0} 1 - \frac{3s+5}{s^2+4s+6} = \frac{1}{6}$$

$$E_{ss} \text{ for } T(s) = \lim_{s \rightarrow 0} s \cdot (1 - T(s)) \cdot R(s)$$

$$= \lim_{s \rightarrow 0} s \left(1 - \frac{3s+5}{s^2+7s+11} \right) \cdot \frac{1}{s}$$

$$= \lim_{s \rightarrow 0} 1 - \frac{3s+5}{s^2+7s+11} = \frac{6}{11}$$

The steady-state error of both $G(s)$ and $T(s)$ are constant and not zero. E_{ss} of $T(s)$ is larger than E_{ss} of $G(s)$

Non-zero constant E_{ss} w.r.t. step input indicates that the system is type 0.

This can also be seen from the TF, that there is no free integrator on the denominator.

e The non-zero constant E_{ss} can be eliminated by adding an integrator (increasing the system type to 1).

However this will make the system aggressive.

Adding a PI controller or a Lag controller is preferable in this case.

5 Given

$$G(s) = \frac{s+3}{(s+2)(s^2+2s+25)} \quad K=2$$

a Bode plot:

$$\begin{aligned} G(s) &= \frac{2(s+3)}{(s+2)(s^2+2s+25)} = \frac{2 \times 3 \left(\frac{s}{3} + 1\right)}{2 \times 25 \left(\frac{s}{2} + 1\right) \left(\left(\frac{s}{5}\right)^2 + \frac{2}{25}s + 1\right)} \\ &= \frac{0.12 \left(\frac{s}{3} + 1\right)}{\left(\frac{s}{2} + 1\right) \left(\left(\frac{s}{5}\right)^2 + \frac{2}{25}s + 1\right)} \end{aligned}$$

Terms:

$$K_0 = 0.12 ; \quad \frac{s}{3} + 1 ; \quad \frac{1}{\frac{s}{2} + 1} ; \quad \frac{1}{\left(\frac{s}{5}\right)^2 + \frac{2}{25}s + 1}$$

See semilog paper

b As the Bode plot is always below 0 for magnitude and never crosses -180° for the phase, then

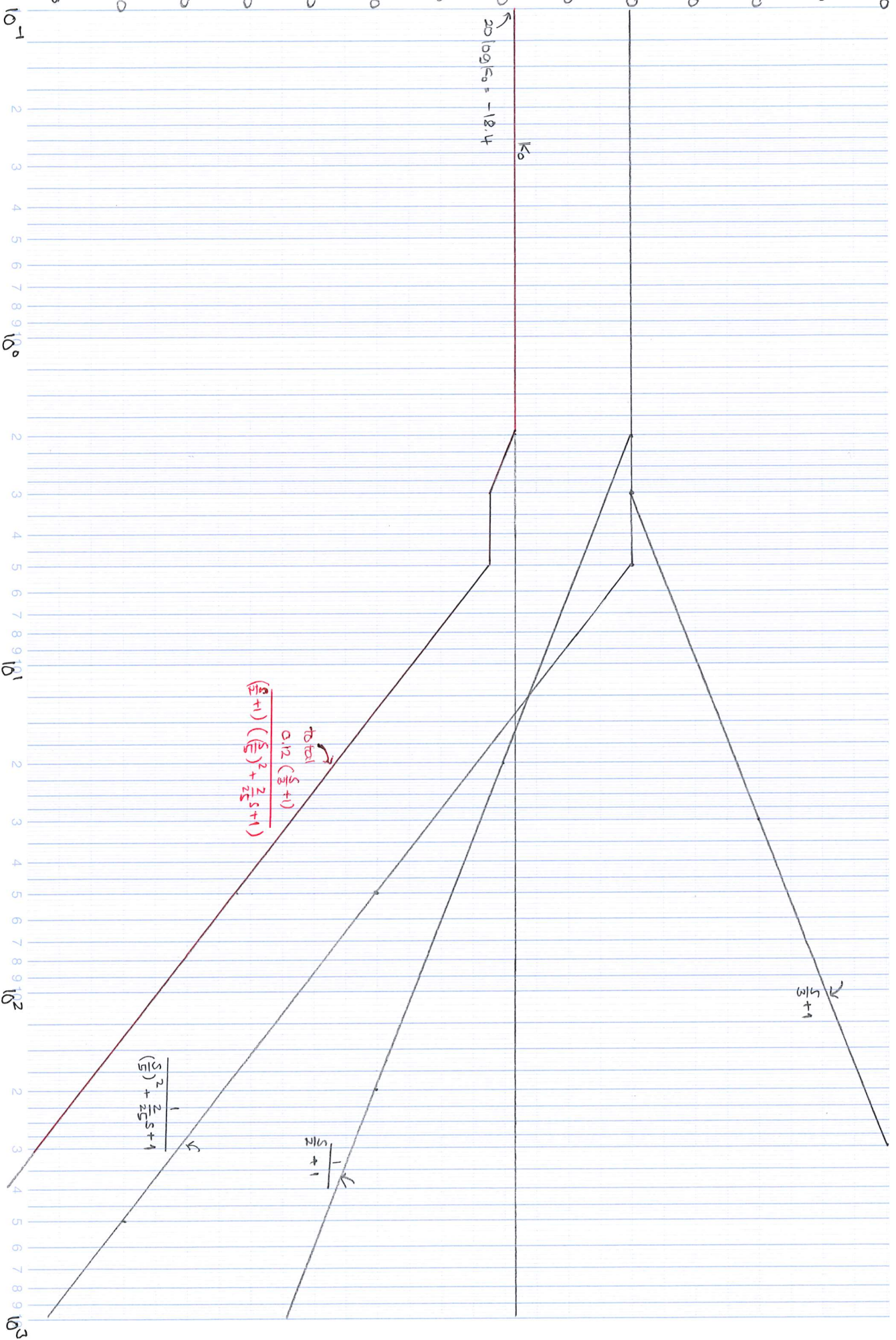
$$GM = \infty, \quad PM = \infty$$

c Both PM and GM are ∞ (positive).

Hence the system is stable. In fact the system is absolutely stable as both PM and GM are infinity.

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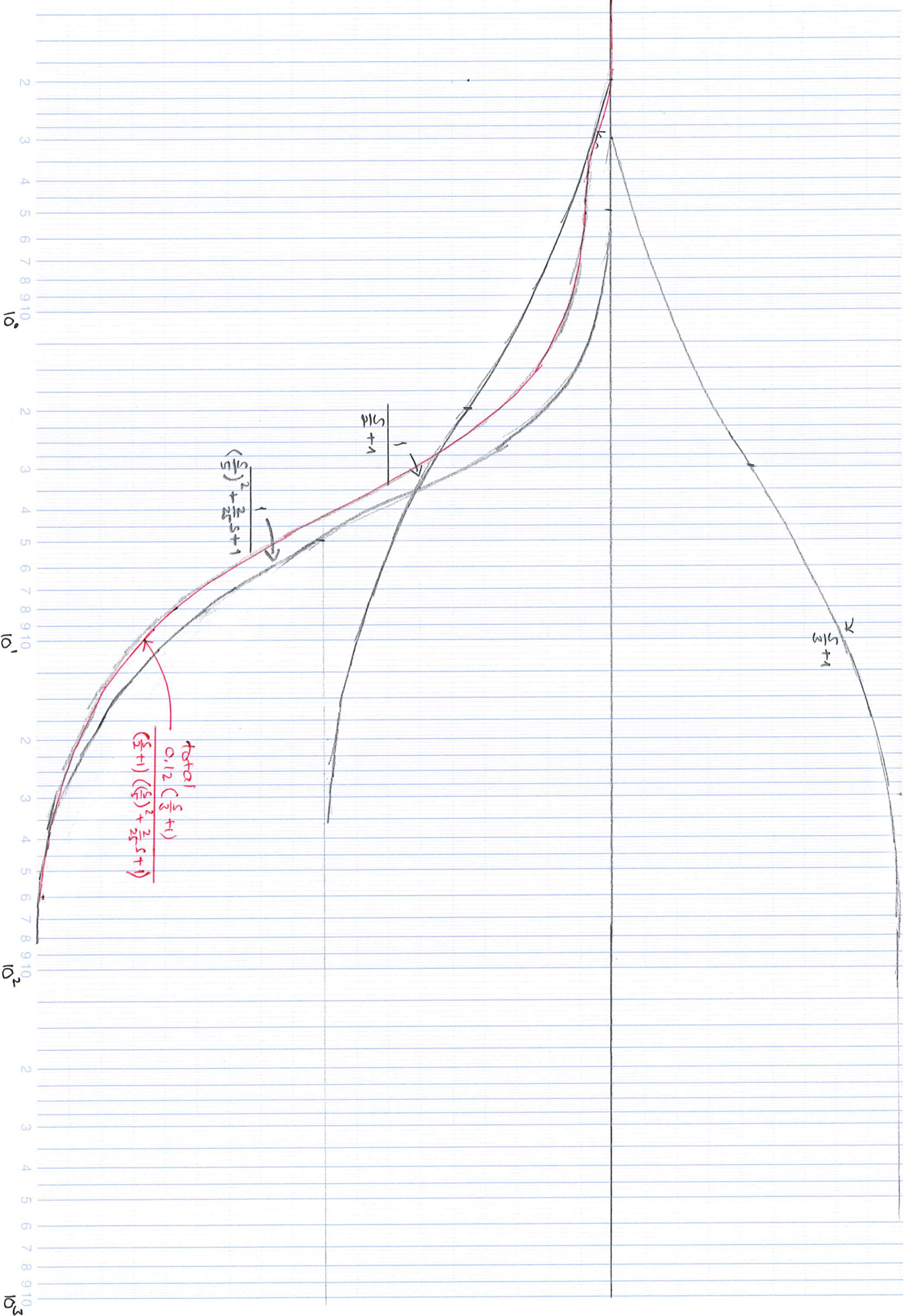
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$L(s, w)$

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6. Open-loop transfer function

$$G(s) = \frac{K(s^2 - 8s + 15)}{s^2 + 2s + 8} \quad K = 1$$

a. $G(j\omega) = \frac{K(j\omega)^2 - 8j\omega + 15}{(j\omega)^2 + 2j\omega + 8} = \frac{(j\omega)^2 - 8j\omega + 15}{(j\omega)^2 + 2j\omega + 8}$ for $K=1$

at $\omega = 0$, $G(j\omega) = \frac{15}{8}$

Hence:

$$|G(j\omega)|_{\omega=0} = \frac{15}{8}, \quad \angle G(j\omega)_{\omega=0} = \tan^{-1} \frac{0}{15/8} = 0^\circ$$

at $\omega \rightarrow \infty$

$$\lim_{\omega \rightarrow \infty} G(j\omega) = \lim_{\omega \rightarrow \infty} \frac{-\omega^2 - 8j\omega + 15}{-\omega^2 + 2j\omega + 8} = 1$$

Hence:

$$|G(j\omega)|_{\omega=\infty} = 1, \quad \angle G(j\omega)_{\omega=\infty} = \frac{0}{1} = 0^\circ \text{ or } 360^\circ$$

b From the Nyquist plot

$N = 2$ clockwise encirclement of $(-1/j0)$

$P = 0$ unstable poles of $G(s)$

$Z = N + P = 2 + 0 = 2$ hence the system is unstable

7 OLTF :

$$G(s) = \frac{6}{s^2 + 2s + 6}$$

a. PD controller : $G_c(s) = K_p + K_d s$

CLTF :

$$T(s) = \frac{G_c(s) G(s)}{1 + G_c(s) G(s)} = \frac{(K_p + K_d s) G(s)}{1 + (K_p + K_d s) G(s)}$$

$$= \frac{6(K_p + K_d s)}{s^2 + 2s + 6 + 6(K_p + K_d s)} = \frac{6(K_p + K_d s)}{s^2 + (2 + 6K_d)s + 6(1 + K_p)}$$

$$T(s) = \frac{6(K_p + K_d s)}{s^2 + (2 + 6K_d)s + 6(1 + K_p)}$$

closed-loop characteristic equation in K_p and K_d :

$$s^2 + (2 + 6K_d)s + 6(1 + K_p) = 0$$

b. 2nd order system parameters ζ , ω_n

$$\frac{s^2}{\omega_n^2} = s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + (2 + 6K_d)s + 6(1 + K_p)$$

Hence :

$$\omega_n^2 = 6(1 + K_p)$$

$$2\zeta\omega_n = 2 + 6K_d$$

$$\omega_n = \sqrt{6(1 + K_p)}$$

$$\zeta\omega_n = 1 + 3K_d \Rightarrow \zeta = \frac{1 + 3K_d}{\sqrt{6(1 + K_p)}}$$

$$c. \omega_n = 8 \Rightarrow \sqrt{6(1 + K_p)} = 8$$

$$\zeta = 0.8 = \frac{1 + 3K_d}{\omega_n} = \frac{1 + 3K_d}{8}$$

$$6(1 + K_p) = 64$$

$$1 + K_p = \frac{32}{3} \Rightarrow K_p = \frac{29}{3}$$

$$6.4 = 1 + 3K_d$$

$$K_d = \frac{5.4}{3} = 1.8$$

d. The effect of PD is adding damping to the system.

(As can be seen, without PD, ω_n for $G(s)$ is $\sqrt{6}$ and $\zeta = \frac{1}{\sqrt{6}} = \frac{1}{\sqrt{6}} \sqrt{6}$

Adding the PD controller increase ζ to 0.8 as in the specs.) = 0.408